SHAMBHAVI

Birla Institute of Technology (BIT), Mesra | Ranchi, India
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PROFESSIONAL SUMMARY

Second-year student at Birla Institute of Technology (BIT), Mesra, with an 8.7 CGPA and a foundation in machine learning, data science, algorithms, and software development. Experience includes product engineering and data science internship work, alongside projects in predictive analytics, explainable AI, and semantic recommendation systems. Seeking AI/ML, data science, or software engineering opportunities.

EDUCATION

Birla Institute of Technology (BIT), Mesra | Ranchi, Jharkhand
Bachelor of Computer Applications | 2025–Present | **CGPA: 8.7/10**

Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Computer Architecture.

TECHNICAL SKILLS

Programming Languages: Python, C, C++, SQL, Java (basic), JavaScript (basic)

Machine Learning and Data Science: Machine Learning, Data Analysis, Data Preprocessing, Feature Engineering, Model Evaluation, Explainable AI

Libraries and Frameworks: Scikit-learn, Pandas, NumPy, Matplotlib, Sentence Transformers, Streamlit, Flask, FastAPI

Web Development and Databases: React.js, Node.js, MongoDB, PostgreSQL

Tools and Core CS: Git, GitHub, Linux (basics), Data Structures and Algorithms, OOP, DBMS, Operating Systems, Computer Networks

EXPERIENCE

Product Engineering & Data Science Intern | Vacation Mantra Technologies Pvt. Ltd.
June 2026 – August 2026 | *Hybrid*

- Contributed to product engineering and data science workflows, supporting application development and data-driven features.
- Worked with Python and software development tools to support data processing, product functionality, and AI-driven initiatives.
- Collaborated on implementation tasks connecting technical requirements with usable product features.

AI/ML Software Engineer Intern | Be11 Sports Technologies Private Limited
Incoming / selected; confirm start date and status before describing as active employment.

- Selected for an AI/ML Software Engineer Intern role focused on technology, AI/ML, and product development initiatives.

PROJECTS

Loan Approval Prediction: ML vs. LLM-Driven Decision System
Python | *Pandas* | *Scikit-learn* | *Flask* | *OpenAI API*

- Developed a comparative loan-approval workflow evaluating traditional models, including Random Forest and Logistic Regression, against an LLM-assisted decision approach.
- Designed preprocessing and feature-engineering pipelines for applicant attributes such as credit history, employment type, and regional factors.
- Used the LendingClub loan dataset as the project data source; confirm the exact subset and evaluation metrics before reporting numeric results.
- Built a Flask interface to compare model predictions and contextual explanations.

Artificial Wisdom System
Python | *Machine Learning* | *Explainable AI* | *Streamlit*

- Developing a research-oriented framework exploring reasoning, adaptive learning, and context-aware decision support.
- Evaluating predictive models and explainability techniques to improve transparency and interpretability.

AI-Driven CRM Analytics System
Python | *Machine Learning* | *React.js* | *Node.js* | *MongoDB*

- Developed an analytics workflow to translate customer-interaction data into business insights.
- Applied preprocessing and exploratory analysis for segmentation, lead scoring, and predictive analytics; designed KPI dashboard views.

Cross-Domain Semantic Recommendation System
Python | *FastAPI* | *React* | *PostgreSQL* | *Sentence Transformers*

- Connected recommendations across movies, books, songs, and games using semantic embeddings and cosine similarity.
- Designed personalization, recommendation match scores, and diversity balancing across content categories.

ONGOING RESEARCH

Context-Aware AI and Explainable Decision Support | Ongoing

- Exploring how predictive machine learning, retrieval-augmented context, and explainability can be combined for more transparent decision support.
- Investigating model evaluation, reliability, and human-centered explanations in applied AI workflows.

CERTIFICATIONS AND COURSEWORK

Stanford OpenCourseWare: CS221 (Artificial Intelligence), CS229 (Machine Learning), CS230 (Deep Learning), CS224N (Natural Language Processing).

Udemy: Data Science, Machine Learning, Deep Learning & NLP Bootcamp by Krish Naik.